

Bharath Reddy Polu

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SUMMARY

Data Scientist with 5+ years of experience in machine learning, statistical analysis, and data visualization to drive data-driven decision-making. Experienced in end-to-end analytics workflows, including data extraction, modeling, predictive analytics, and reporting. Skilled in leveraging cloud technologies and automation to optimize business processes and enhance strategic insights. Adept at translating complex data into actionable recommendations to support operational and financial objectives.

WORK EXPERIENCE

MetLife

Aug 2024 - Current

Data Scientist

- Developed a predictive risk scoring model using XGBoost, improving claim risk assessment accuracy by **15%** and supporting an initial cost reduction of **\$250K**
- Built financial performance forecasting models with AWS SageMaker, enhancing portfolio risk prediction accuracy by **20%** across two key financial product lines
- Conducted sentiment analysis on **18,000+** customer communication records, uncovering five key areas for customer experience improvement
- Designed four interactive tableau dashboards for various product reporting use cases, reducing manual reporting time by **35%** and improving data accessibility.
- Developed and optimized CI/CD pipelines for machine learning model deployment, automating the integration process and reducing deployment time by **25%**.

Neon IT Systems

Jan 2019 - Jun 2023

Senior Data Science Analyst

- Developed a fraud detection model using Random Forest, reducing fraudulent transactions by 20% and saving the company **\$50K** per month
- Developed risk assessment models using SVM, Random Forest, and Gradient Boosting, improving financial risk management KPIs by **12%** and streamlining operational processes for better decision-making
- Designed and optimized data processing pipelines, reducing ingestion time by **35%** and contributing to **\$87K** in cost saving
- Applied machine learning A/B testing to improve digital financial products, increasing conversion rates by 15% and generating **\$340K** in additional revenue. -keep either this or last one
- Created interactive Power BI and tableau dashboards, cutting reporting cycle time by **30%** and improving financial forecasting accuracy for executives.

SKILLS

- Machine Learning & Statistical Modeling:** Regression Analysis, Bayesian Method, Decision Tree, Random Forests, Neural Network, Sentiment Analysis, K-Means Clustering, KNN, Classification, SVM, Natural Language Processing (NLP), LLM,CNN, XGBoost
- Data Engineering & Cloud Technologies:** AWS SageMaker, Azure Data Bricks, AWS Glue, Data Cleaning, Data Wrangling, Databricks, Big Query, Snowflake, Docker, Kubernetes, ETL
- Programming Languages & Tools:** Python, R, SQL, SAS, NumPy, Pandas, Matplotlib, SciPy, Scikit-Learn, PyTorch, TensorFlow, Keras, Spark
- Data Visualization & Business Intelligence:** Tableau, Power BI, Microsoft Excel, MySQL, SQL Server, Oracle, MongoDB, JMP

EDUCATION

University of Cincinnati - Cincinnati, Ohio

Aug - 2024

Master of Science, Business Analytics (STEM), Dept. of Data Sciences and Operations: **GPA: 3.8/4**

ACADEMIC PROJECTS

[Real-time fraud detection with machine learning model](#) : Python (NumPy, Pandas, Matplotlib, sci-kit), Tableau

- Developed a real-time fraud detection algorithm utilizing synthetic credit applications data, optimizing a Neural Network model with a **0.735** score at **3%** Fraud Detection Rate (FDR)
- Analyzed and preprocessed data to address quality issues, performed imputation for missing values, and standardized variables using Z-scaling to improve data consistency for modeling
- Feature engineered additional **421** variables to increase fraud signals in the data; Performed feature selection using Recursive feature elimination (RFECV) to identify significant variables, reducing the number of predictor variables from **435 to 30**